

Bathroom Concept: 5. Heating cable on the membrane

Bathroom renovation · Bathroom (no site address recorded) · pack 05 of 9 · printed 2026-10-08 · 11 A3 pages

PROPOSED · FOR TRADE REVIEW · NOT AS-BUILT · NOT A COMPLIANCE CERTIFICATE

The proposed in-screed heating cable route, laid by the owner and tested by the electrician before the screed. The specification sheet is the compact one-page form: route points and cable levels on one row each. Proposed, pending manufacturer and electrician review.

For: Electrician.

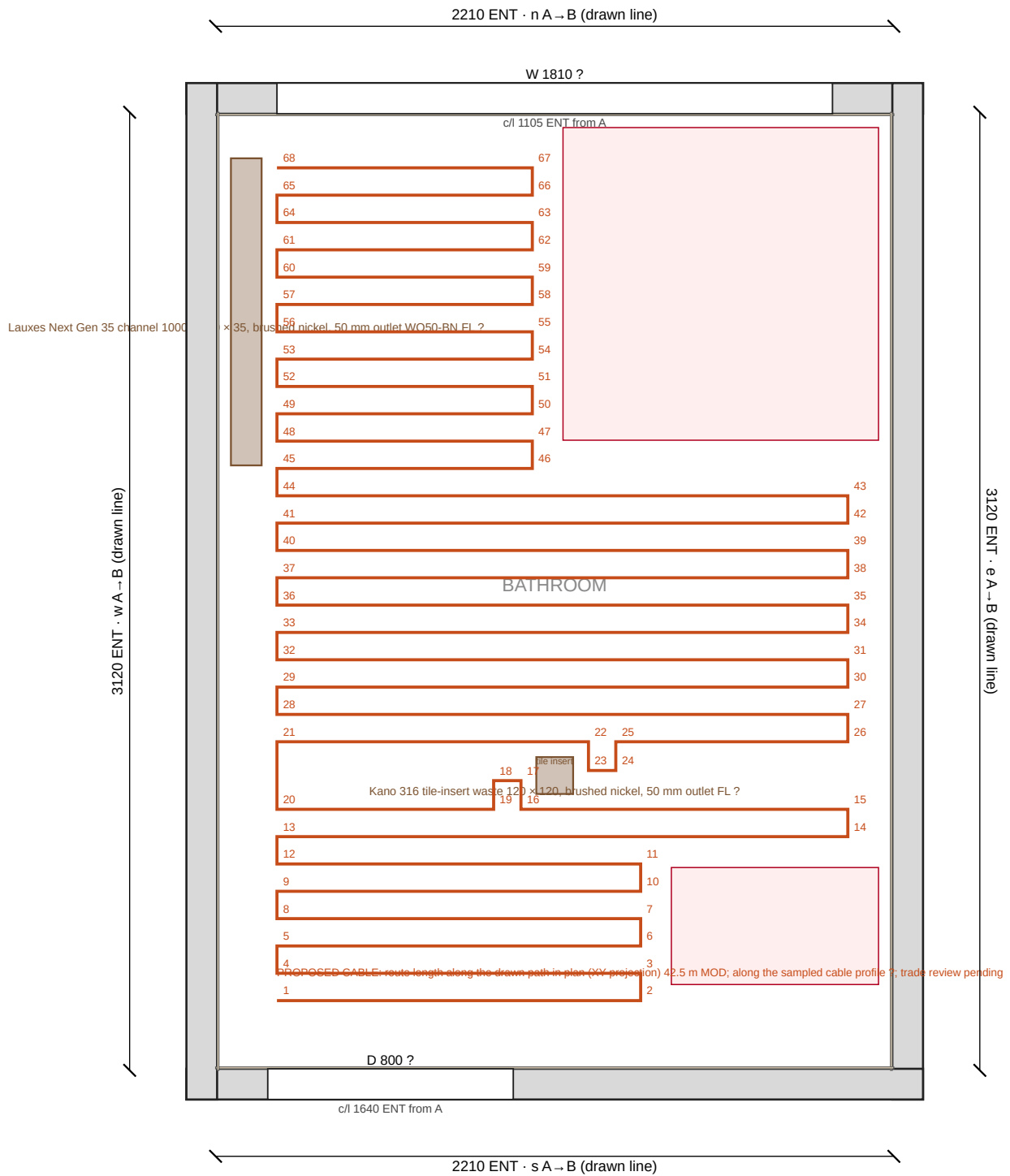
How to read these sheets

- Dimensions in millimetres. Every figure carries its status: SC site-confirmed · M measured · PUB published · P proposed · E estimated · DER derived · ENT entered · DEF default · ? unknown. Treat anything not SC or M as a figure to check on site.
- Elevations look at a wall from inside the room; end A is on the left. Wall fittings are set out from the wall's finished (tile) face: centreline along from the return wall's face at A, bottom above the finished floor.
- F1, F2... are the fixtures (fixture schedule in 00-overview.pdf); F11.2 is service point 2 of fixture 11. "?" means not known yet and never drawn.
- Scale 1:20 at A3 on the plans and elevations. Print at 100% ("actual size"), never "fit to page".
- A wall elevation is left out when nothing new is set out on that wall at this stage; the plan and specification still cover the stage. Open decisions and what is still to measure: 00-overview.pdf.

Generated from the project model by npm run printables; the drawings are the same exports saved in the repository (shots/stage-pack, shots/sample-sheets).

In this pack

Sheet	Page
Plan	2
Specification	3
Heating review	4–11



Stage: 5. Heating cable on the membrane

Shows 16 element(s) of the one project model; everything else is hidden, not removed.
Status: SC site-confirmed · M measured · PUB published · P proposed · E estimated · DER derived · ENT entered · DEF default · ? unknown
Not modelled, never drawn: pipe and cable runs between service points.

Wall faces and layers (mm from drawn line)

Wall n right: Villaboard 6 mm: thickness 6 P · outer face position 11 E
Wall e right: Villaboard 6 mm: thickness 6 P · outer face position 11 E
Wall s right: Villaboard 6 mm: thickness 6 P · outer face position 11 E
Wall w right: Villaboard 6 mm: thickness 6 P · outer face position 11 E

Floor (mm above datum)

Bathroom floor: substrate concrete slab, framed in place (house on piers) ENT · substrate top level -120 E · finished level target 0 P · Tiler's screed (heating cable inside) + Tiler's adhesive together, to reach the target 108.5 E
Bathroom floor: Membrane (Bastion liquid polyurethane): thickness 1.5 E · top level -118.5 E

Drainage / heating

Bathroom: proposed heating cable: manufacturer ? · model SCK0765L ENT · product length (m) 42.5 PUB · rated output (W) 765 PUB · derived spacing min 87.1 DER · derived spacing max 120 DER · plan route length (m) 42.5 MOD · spatial route length, sampled profile (m) ?
Bathroom: Lauxes Next Gen 35 channel 1000 × 100 × 35, brushed nickel, 50 mm outlet WO50-BN-FL ? · finished level at waste ? · drain product Lauxes Next Gen 35 Custom 1000, brushed nickel NAM · grate 1000 × 100 M
Bathroom: Kano 316 tile-insert waste 120 × 120, brushed nickel, 50 mm outlet: finished level at waste ? · drain product Kano 316 Tile Insert Waste 120×120, brushed nickel NAM · grate 120 × 120 tile insert M

Unresolved in this view (15)

- Bathroom: proposed heating cable: heating_signoff: Proposed route only: manufacturer and licensed electrician must review cable identity, length, output, bend radius, spacing, exclusions, cover, sensor, cold tails, waterproofing and electrical installation. No compliance approval. No electrical approval is implied.
- Bathroom: proposed heating cable: heating_product_unresolved: sample-sck0765l-carton is not an accepted heating-cable in this library. Checks use the travelling snapshot only when its productId is that id or in its series.
- Bathroom: proposed heating cable: heating_product_unresolved: sample-mwd5-1999-cbp3 is not an accepted thermostat in this library. Checks use the travelling snapshot only when its productId is that id or in its series.
- Bathroom: proposed heating cable: heating_metadata_unknown: Unknown cable/product information: manufacturer, minSpacing, depthFromBottom.
- Bathroom: proposed heating cable: heating_route_length_unknown: Spatial cable length and confirmed product-length balance remain unknown until the whole route has resolved screed levels and cable height; the plan projection is modelled, not cable length.
- Bathroom: proposed heating cable: heating_ip_location: Thermostat printed IP21 (published, carton label — Photographed MWD5-1999-CBP3 carton label: housing IP21) vs planned location outside a wet room (Hallway wall outside the bathroom, next to the light switch (right as you look into the bathroom), 1000 mm off the floor. Not drawn on this plan.; Owner, 5 Oct 2026; height raised to 1000 mm, 7 Oct 2026). This check does not decide zone suitability; the electrician decides. No compliance approval.
- Bathroom: proposed heating cable: heating_depth_unknown: Cable section remains unresolved until screed layer, floor levels and cable height (bottom face of the selected screed layer (top of the layer below / subfloor stack); not the underside of the tile unless that face is the screed top) are entered.
- Bathroom: proposed heating cable: manufacturer unknown; printed as "?" (missing manufacturer).

... 7 more: see the specification sheet

Project: Bathroom Concept

Site / room: Bathroom (no site address recorded)

Stage diagram: 5. Heating cable on the membrane

Scale 1:20 @ A3 · Exported 2026-10-08 · not a revision of A-01

Prepared by: ? · Plan: Bathroom Concept

PROPOSED · FOR TRADE REVIEW · NOT AS-BUILT · NOT A COMPLIANCE CERTIFICATE

Specification: 5. Heating cable on the membrane

Project: Bathroom Concept · Site / room: Bathroom (no site address recorded) · Prepared by: ? · Plan: Bathroom Concept
Exported 2026-10-08 · not a revision of A-01

PROPOSED · FOR TRADE REVIEW · NOT AS-BUILT · NOT A COMPLIANCE CERTIFICATE

Lists exactly the 16 element(s) visible in this stage view of the one project model. Lengths in mm. Status: SC site-confirmed · M measured · PUB published · P proposed · E estimated · DER derived (converted, not published) · ENT entered (not site-confirmed) · DEF default placeholder · ? unknown. A "?" value is not known and must not be read as a measurement. Not modelled, so never listed: pipe and cable runs between service points (only the points are modelled).

Property	Value	Status	Measured from	Source	Missing
Walls as drawn (line, length, body)					
Wall n					
length A – B along drawn line (mm)	2210	ENT entered			
thickness (mm)	100	? unknown			
height (mm)	2700	? unknown			
Wall e					
length A – B along drawn line (mm)	3120	ENT entered			
thickness (mm)	100	? unknown			
height (mm)	2700	? unknown			
Wall s					
length A – B along drawn line (mm)	2210	ENT entered			
thickness (mm)	100	? unknown			
height (mm)	2700	? unknown			
Wall w					
length A – B along drawn line (mm)	3120	ENT entered			
thickness (mm)	100	? unknown			
height (mm)	2700	? unknown			
Wall board (e.g. Villaboard)					
Wall n right: Villaboard 6 mm					
kind	board	ENT entered			
thickness (mm)	6	P proposed		[1]	
outer face position (mm)	11	E estimated	[2]		
Wall e right: Villaboard 6 mm					
kind	board	ENT entered			
thickness (mm)	6	P proposed		[1]	
outer face position (mm)	11	E estimated	[3]		
Wall s right: Villaboard 6 mm					
kind	board	ENT entered			
thickness (mm)	6	P proposed		[1]	
outer face position (mm)	11	E estimated	[4]		
Wall w right: Villaboard 6 mm					
kind	board	ENT entered			
thickness (mm)	6	P proposed		[1]	
outer face position (mm)	11	E estimated	[5]		
Windows					
Window window_n on wall_n					

References (47): full text of the cells printed as [n] above

- owner's tile and build-up choices
- drawn line of wall_n, toward right side
- drawn line of wall_e, toward right side
- drawn line of wall_s, toward right side
- drawn line of wall_w, toward right side
- wall_n end A, along the drawn line
- wall_s end A, along the drawn line
- concrete slab, framed in place (house on piers)
- existing floor surface (current tile top)
- owner: about 120 mm below the current tile, seen from underneath; confirm after demolition
- finished tile back at the current tile level, so the doorway stays flush (to confirm)
- Tiler's screed (heating cable inside) + Tiler's adhesive together, to reach the target (mm)
- owner, 7 Oct 2026: liquid polyurethane, about 1.5 mm
- product identity (submission text, not a published figure)
- Owner lays it on the cured membrane, before the tiler's screed, fixed with Ametalin tape (polyurethane membrane; owner holds 100 m) at the run ends and every 1–1.5 m, per the Coldbuster in-screed manual (July 2023, taken as this cable's manual: it does not name SCK0765L). The electrician tests it before and after the screed (the manual's p. 16 record, for the warranty), wires the thermostat on an RCD circuit, and runs the floor sensor's conduit. Owner, 5 Oct 2026: the thermostat end is at the bathroom doorway, where the floor beyond is timber; the 3.0 m cold tail (manual; the manual's cable is Ø6 mm) goes down through it

Unresolved in this view (15)

- Bathroom: proposed heating cable: heating_signoff: Proposed route only: manufacturer and licensed electrician must review cable identity, length, output, bend radius, spacing, exclusions, cover, sensor, cold tails, waterproofing and electrical installation. No compliance approval. No electrical approval is implied.
- Bathroom: proposed heating cable: heating_product_unresolved: sample-sck0765l-carton is not an accepted heating-cable in this library. Checks use the travelling snapshot only when its productid is that id or in its series.
- Bathroom: proposed heating cable: heating_product_unresolved: sample-mwd5-1999-cbp3 is not an accepted thermostat in this library. Checks use the traveling snapshot only when its productid is that id or in its series.
- Bathroom: proposed heating cable: heating_metadata_unknown: Unknown cable/product information: manufacturer, minSpacing, depthFromBottom.

- to under the house (on piers). Owner, 7 Oct 2026: the cable runs under the shower (the electrician approved it) but not under the bath, so the heated area stays within the 5.1 m² that 42.5 m covers. Owner, 8 Oct 2026: no cable under the toilet (its pan is screwed to the floor through brackets); the cable does run under the vanity. Proposed loop: east–west runs, 100 mm off the walls, the bath and the toilet (manual, p. 6), 150 mm off the left wall to clear the linear drain, routed round the Kano waste (manual, p. 10); 89 mm throughout (owner, 8 Oct 2026: the cable kept clear of the toilet goes into closer runs and round the Kano waste), inside the derived coverage/length range (3.7–5.1 m² / 42.5 m = 87–120 mm) and the Coldbuster manual's 70–120 mm for bathrooms (p. 7); two hooks beside the Kano waste take up the rest so the loop uses exactly 42.5 m. Modelling choice, not a manufacturer or code spacing requirement. Plan length is exactly 42.5 m, equal to the published heated length (zero slack); any resolved fail makes the spatial route exceed 42.5 m once screed levels resolve.
- carton label — SCK0765L in-screed heating cable carton: heated length 42.5 m
 - carton label — SCK0765L in-screed heating cable carton: 765 W
 - carton label — SCK0765L in-screed heating cable carton: coverage 3.7–5.1 m², minimum
 - carton label — SCK0765L in-screed heating cable carton: coverage 3.7–5.1 m², maximum
 - coverage area / cable length - Implied loop spacing if the published coverage range is spread along the published heated length. Derived, not a manufacturer spacing instruction and not a code requirement. Inputs: coverage area, minimum 3.7 m² (published, carton label — SCK0765L in-screed heating cable carton: coverage 3.7–5.1 m², minimum) and heated cable length 42.5 m (published, carton label — SCK0765L in-screed heating cable carton: heated length 42.5 m).

- coverage area / cable length · Implied loop spacing if the published coverage range is spread along the published heated length. Derived, not a manufacturer spacing instruction and not a code requirement. Inputs: coverage area, maximum 5.1 m² (published, carton label — SCK0765L in-screed heating cable carton: coverage 3.7–5.1 m², maximum) and heated cable length 42.5 m (published, carton label — SCK0765L in-screed heating cable carton: heated length 42.5 m).
- carton label — SCK0765L in-screed heating cable carton: 3.2 A
- carton label — SCK0765L in-screed heating cable carton: 240 V AC
- carton label — Photographed MWD5-1999-CBP3 carton label: 16 A
- carton label — Photographed MWD5-1999-CBP3 carton label: 100–240 V AC
- carton label — Photographed MWD5-1999-CBP3 carton label: housing IP21
- minSpacing (mm)
- Coldbuster In Screed Heater Kit Installation Manual (July 2023), p. 6: mark 100–200 mm from the walls; Coldbuster recommends 100 mm. Taken as this cable's manual: it does not name SCK0765L
- depthFromBottom (mm)
- bottom face of the selected screed layer (top of the layer below / subfloor stack); not the underside of the tile unless that face is the screed top
- selected zone boundary as stored (sample room rectangle = existing internal surface). Intended trade datum is the finished wall face; that conversion is not applied.
- XY projection of the drawn path. Modelled, not published cable length.

- Sampled cable profile. Modelled; unknown until screed levels and cable height resolve. Not published cable length.
- confirmed product length – modelled spatial route length · Balance stays unknown until confirmed product length and the whole spatial profile are both numeric.
- Balance stays unknown until confirmed product length and the whole spatial profile are both numeric.
- Plan route length is the XY projection (modelled). Spatial route length follows the sampled cable profile (modelled) and drives confirmed product balance only when the whole profile resolves. Product length comes from the referenced brief. Unsupplied cold tails and connections are excluded.
- coverage area / cable length: Implied loop spacing if the published coverage range is spread along the published heated length. Derived, not a manufacturer spacing instruction and not a code requirement.
- Pending manufacturer / electrician review. No electrical or compliance approval.
- 1: 150 / 2843 P; 2: 1335 / 2843 P; 3: 1335 / 2754 P; 4: 150 / 2754 P; 5: 150 / 2665 P; 6: 1335 / 2665 P; 7: 1335 / 2576 P; 8: 150 / 2576 P; 9: 150 / 2487 P; 10: 1335 / 2487 P; 11: 1335 / 2398 P; 12: 150 / 2398 P; 13: 150 / 2309 P; 14: 2010 / 2309 P; 15: 2010 / 2220 P; 16: 945 / 2220 P; 17: 945 / 2126.5 P; 18: 856 / 2126.5 P; 19: 856 / 2220 P; 20: 150 / 2220 P; 21: 150 / 2000 P; 22: 1165 / 2000 P; 23: 1165 / 2093.5 P; 24: 1254 / 2093.5 P; 25: 1254 / 2000 P; 26: 2010 / 2000 P; 27: 2010 / 1911 P; 28: 150 / 1911 P; 29: 150 / 1822 P; 30: 2010 / 1822 P; 31: 2010 / 1733 P; 32: 150 / 1733 P; 33: 150 / 1644 P; 34: 2010 / 1644 P; 35: 2010 / 1555 P; 36: 150 / 1555 P; 37: 150 / 1466 P; 38: 2010 / 1466 P; 39: 2010 / 1377 P; 40: 150 / 1377 P; 41: 150 / 1288 P; 42: 2010 / 1288 P; 43: 2010 / 1199 P; 44: 150 / 1199 P; 45: 150 / 1110 P; 46: 982 / 1110 P; 47: 982 / 1021 P; 48: 150 / 1021 P; 49: 150 / 932 P; 50: 982 / 932 P; 51: 982 / 843 P; 52: 150 / 843 P; 53:

- 150 / 754 P; 54: 982 / 754 P; 55: 982 / 665 P; 56: 150 / 665 P; 57: 150 / 576 P; 58: 982 / 576 P; 59: 982 / 487 P; 60: 150 / 487 P; 61: 150 / 398 P; 62: 982 / 398 P; 63: 982 / 309 P; 64: 150 / 309 P; 65: 150 / 220 P; 66: 982 / 220 P; 67: 982 / 131 P; 68: 150 / 131 P
- cable level at 197 sampled points along plan route (mm)
- plane "Dry area" is unresolved (missing Kano 316 tile-insert waste 120 × 120, brushed nickel, 50 mm outlet level, fall or a control level); cable height above screed bottom, Tiler's screed (heating cable inside) thickness, Tiler's adhesive thickness, plane "Beside the shower (under the bath)" is unresolved (missing Kano 316 tile-insert waste 120 × 120, brushed nickel, 50 mm outlet level, fall or a control level), plane "Shower" is unresolved (missing Lauxes Next Gen 35 channel 1000 × 100 × 35, brushed nickel, 50 mm outlet WO50-BN level, fall or a control level)
- Owner, 7 Oct 2026: floor under the bath left unheated so the heated area stays within the 5.1 m² the 42.5 m cable covers
- Owner, 8 Oct 2026: no cable under the toilet (its pan is screwed to the floor through brackets); cable stays under the vanity
- heating_signoff, heating_product_unresolved, heating_product_unresolved, heating_metadata_unknown, heating_route_length_unknown, heating_ip_location, heating_depth_unknown
- finished level at waste (mm)
- Lauxes Next Gen 35 Custom 1000, brushed nickel
- Kano 316 Tile Insert Waste 120×120, brushed nickel

- height above screed bottom, Tiler's screed (heating cable inside) thickness, Tiler's adhesive thickness, plane "Beside the shower (under the bath)" is unresolved (missing Kano 316 tile-insert waste 120 × 120, brushed nickel, 50 mm outlet level, fall or a control level), plane "Shower" is unresolved (missing Lauxes Next Gen 35 channel 1000 × 100 × 35, brushed nickel, 50 mm outlet WO50-BN level, fall or a control level).
- Bathroom: Lauxes Next Gen 35 channel 1000 × 100 × 35, brushed nickel, 50 mm outlet WO50-BN: finished level at waste (mm) unknown; printed as "?" (missing finished level at waste (mm)).
 - Bathroom: Kano 316 tile-insert waste 120 × 120, brushed nickel, 50 mm outlet: finished level at waste (mm) unknown; printed as "?" (missing finished level at waste (mm)).

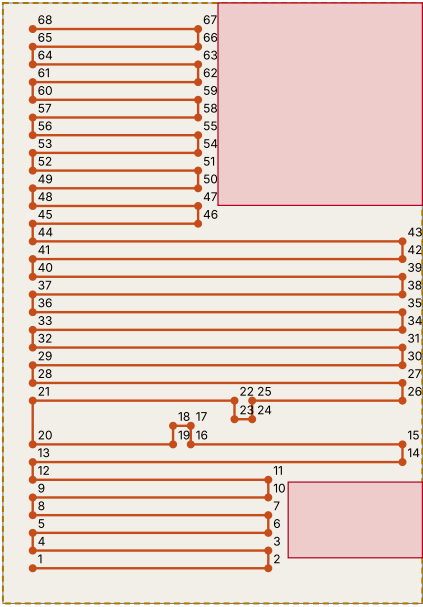
Property	Value	Status	Measured from	Source	Missing
width (mm)	1810	? unknown			
sill above floor (mm)	1520	? unknown			
height (mm)	600	? unknown			
centre from end A (mm)	1105	ENT entered	[6]		
Doors					
Door door_s on wall_s					
width (mm)	800	? unknown			
height (mm)	1900	? unknown			
centre from end A (mm)	1640	ENT entered	[7]		
Rooms (outline label)					
Room Bathroom					
size w × h (mm)	2110 × 3020	ENT entered			
Floor substrate and datum					
Bathroom floor: substrate					
substrate	[8]	ENT entered			
substrate top level (mm)	-120	E estimated	[9]	[10]	
finished level target (mm)	0	P proposed	[9]	[11]	
[12]	108.5	E estimated	substrate top to target		
Floor waterproofing					
Bathroom floor: Membrane (Bastion liquid polyurethane)					
kind	waterproofing	ENT entered			
thickness (mm)	1.5	E estimated		[13]	
top level (mm)	-118.5	E estimated	[9]		
Proposed in-screed heating cable					
Bathroom: proposed heating cable					
manufacturer	?	? unknown			manufacturer
model	SCK0765L	ENT entered		[14]	
productSource	SCK0765L carton label	ENT entered			
requirements	[15]	ENT entered			
product length (m)	42.5	PUB published	fixture-end	[16]	
rated output (W)	765	PUB published		[17]	
coverage min (m²)	3.7	PUB published		[18]	
coverage max (m²)	5.1	PUB published		[19]	
derived spacing min (mm)	87.1	DER derived		[20]	
derived spacing max (mm)	120	DER derived		[21]	
cable rated current (A)	3.2	PUB published		[22]	

Property	Value	Status	Measured from	Source	Missing
cable rated voltage (V)	240	PUB published		[23]	
thermostat switching current (A)	16	PUB published		[24]	
thermostat voltage min (V)	100	PUB published		[25]	
thermostat voltage max (V)	240	PUB published		[25]	
thermostat printed IP	IP21	PUB published		[26]	
minSpacing (mm)	?	? unknown			[27]
edgeClearance (mm)	100	PUB published		[28]	
depthFromBottom (mm)	?	? unknown			[29]
cable depth datum	[30]	NAM named			
wall setback datum	[31]	NAM named			
plan route length (m)	42.5	MOD modelled		[32]	
spatial route length, sampled profile (m)	?	? unknown		[33]	[33]
remaining confirmed product length (m)	?	? unknown		[34]	[35]
length basis	[36]	P proposed			
spacing formula	[37]	DER derived			
zone ids	bathroom	P proposed			
available zone area (m²), not heat coverage	5.0685	P proposed			
mininum non-adjacent spacing (mm)	89	MOD modelled			
installation approval	[38]	P proposed			
route points 1–68 x / y (mm)	[39]	P proposed	plan origin		
[40]	?	? unknown	[9]		[41]
keep-out Bath x / y / w / h (mm)	1082 / 0 / 1028 / 1018	ENT entered		[42]	
keep-out Toilet x / y / w / h (mm)	1435 / 2409.5 / 675 / 381	ENT entered		[43]	
open item codes (see Unresolved)	[44]	P proposed			

Floor wastes					
Bathroom: Lauxes Next Gen 35 channel 1000 × 100 × 35, brushed nickel, 50 mm outlet WO50-BN					
kind	linear	ENT entered			
finished level at waste (mm)	?	? unknown	[9]		[45]
drain product	[46]	NAM named			
grate (mm)	1000 × 100	M measured			
Bathroom: Kano 316 tile-insert waste 120 × 120, brushed nickel, 50 mm outlet					
kind	point	ENT entered			
finished level at waste (mm)	?	? unknown	[9]		[45]
drain product	[47]	NAM named			
grate (mm)	120 × 120 tile insert	M measured			

Bathroom · proposed in-screed heating

Planning and communication only. Manufacturer / electrician approval pending; not an electrical installation or compliance approval.



Plan route length: 42.5 m · Spatial route length (sampled profile): unknown · selected zone footprint: 6.3722 m² · zone area excluding entered keep-outs: 5.0685 m². Coverage-range check uses availableArea (zone footprints excluding entered keep-outs), the same resolver as evidence. That area is not verified heat coverage. Plan route length is the XY projection (modelled). Spatial route length follows the sampled cable profile (modelled) and drives confirmed product balance only when the whole profile resolves. Product length comes from the referenced brief. Unsupplied cold tails and connections are excluded.

Minimum non-adjacent spacing: 89 mm · boundary clearance: 100 mm · keep-out clearance: 100 mm · remaining confirmed cable length: unknown.

Cable depth datum: bottom face of the selected screed layer (top of the layer below / subfloor stack); not the underside of the tile unless that face is the screed top. Wall setback datum: selected zone boundary as stored (sample room rectangle = existing internal surface). Intended trade datum is the finished wall face; that conversion is not applied.. coverage area / cable length: Implied loop spacing if the published coverage range is spread along the published heated length. Derived, not a manufacturer spacing instruction and not a code requirement.

Product / constraint	Value	Kind / source
manufacturer	unknown	unknown
model	SCK0765L	product identity (submission text, not a published figure)
productSource	SCK0765L carton label	User-entered text
requirements	Owner lays it on the cured membrane, before the tiler's screed, fixed with Ametalin tape (polyurethane membrane; owner holds 100 m) at the run ends and every 1–1.5 m, per the Coldbuster in-screed manual (July 2023, taken as this cable's manual: it does not name SCK0765L). The electrician tests it before and after the screed (the manual's p. 16 record, for the warranty), wires the thermostat on an RCD circuit, and runs the floor sensor's conduit. Owner, 5 Oct 2026: the thermostat end is at the bathroom doorway, where the floor beyond is timber; the 3.0 m cold tail (manual; the manual's cable is Ø6 mm) goes down through it to under the house (on piers). Owner, 7 Oct 2026: the cable runs under the shower (the electrician approved it) but not under the bath, so the heated area stays within the 5.1 m² that 42.5 m covers. Owner, 8 Oct 2026: no cable under the toilet (its pan is screwed to the floor through brackets); the cable does run under the vanity. Proposed loop: east–west runs, 100 mm off the walls, the bath and the toilet (manual, p. 6), 150 mm off the left wall to clear the linear drain, routed round the Kano waste (manual, p. 10); 89 mm throughout (owner, 8 Oct 2026: the cable kept clear of the toilet goes into closer runs and round the Kano waste), inside the derived coverage/length range (3.7–5.1 m² / 42.5 m ≈ 87–120 mm) and the Coldbuster manual's 70–120 mm for bathrooms (p. 7); two hooks beside the Kano waste take up the rest so the loop uses exactly 42.5 m. Modelling choice, not a manufacturer or code spacing requirement. Plan length is exactly 42.5 m, equal to the published heated length (zero slack): any resolved fall makes the spatial route exceed 42.5 m once screed levels resolve.	User-entered text
screedLayerId	floor_screed	User-entered text
Cable product length	42.5 m · published · carton label — SCK0765L in-screed heating cable carton: heated length 42.5 m	published · project-snapshot (unresolved) · carton label — SCK0765L in-screed heating cable carton: heated length 42.5 m · SCK0765L in-screed heating cable carton: heated length 42.5 m
Rated output	765 W · published · carton label — SCK0765L in-screed heating cable carton: 765 W	published · project-snapshot (unresolved) · carton label — SCK0765L in-screed heating cable carton: 765 W · SCK0765L in-screed heating cable carton: 765 W
Coverage area, minimum	3.7 m² · published · carton label — SCK0765L in-screed heating cable carton: coverage 3.7–5.1 m², minimum	published · project-snapshot (unresolved) · carton label — SCK0765L in-screed heating cable carton: coverage 3.7–5.1 m², minimum · SCK0765L in-screed heating cable carton: coverage 3.7–5.1 m², minimum
Coverage area, maximum	5.1 m² · published · carton label — SCK0765L in-screed heating cable carton: coverage 3.7–5.1 m², maximum	published · project-snapshot (unresolved) · carton label — SCK0765L in-screed heating cable carton: coverage 3.7–5.1 m², maximum · SCK0765L in-screed heating cable carton: coverage 3.7–5.1 m², maximum

Derive d spaci ng, minim um	0.0871 m · derived · coverage area / cable length · Implied loop spacing if the published coverage range is spread along the published heated length. Derived, not a manufacturer spacing instruction and not a code requirement. Inputs: coverage area, minimum 3.7 m ² (published, carton label — SCK0765L in-screed heating cable carton: coverage 3.7–5.1 m ² , minimum) and heated cable length 42.5 m (published, carton label — SCK0765L in-screed heating cable carton: heated length 42.5 m).	derived · coverage area / cable length · coverage-area / cable-length · Implied loop spacing if the published coverage range is spread along the published heated length. Derived, not a manufacturer spacing instruction and not a code requirement. Inputs: coverage area, minimum 3.7 m ² (published, carton label — SCK0765L in-screed heating cable carton: coverage 3.7–5.1 m ² , minimum) and heated cable length 42.5 m (published, carton label — SCK0765L in-screed heating cable carton: heated length 42.5 m).
Derive d spaci ng, maxi mum	0.12 m · derived · coverage area / cable length · Implied loop spacing if the published coverage range is spread along the published heated length. Derived, not a manufacturer spacing instruction and not a code requirement. Inputs: coverage area, maximum 5.1 m ² (published, carton label — SCK0765L in-screed heating cable carton: coverage 3.7–5.1 m ² , maximum) and heated cable length 42.5 m (published, carton label — SCK0765L in-screed heating cable carton: heated length 42.5 m).	derived · coverage area / cable length · coverage-area / cable-length · Implied loop spacing if the published coverage range is spread along the published heated length. Derived, not a manufacturer spacing instruction and not a code requirement. Inputs: coverage area, maximum 5.1 m ² (published, carton label — SCK0765L in-screed heating cable carton: coverage 3.7–5.1 m ² , maximum) and heated cable length 42.5 m (published, carton label — SCK0765L in-screed heating cable carton: heated length 42.5 m).
Cable rated curre nt	3.2 A · published · carton label — SCK0765L in-screed heating cable carton: 3.2 A	published · project-snapshot (unresolved) · carton label — SCK0765L in-screed heating cable carton: 3.2 A · SCK0765L in-screed heating cable carton: 3.2 A
Cable rated voltag e	240 V · published · carton label — SCK0765L in-screed heating cable carton: 240 V AC	published · project-snapshot (unresolved) · carton label — SCK0765L in-screed heating cable carton: 240 V AC · SCK0765L in-screed heating cable carton: 240 V AC
Ther mosta t rated switch ing curre nt	16 A · published · carton label — Photographed MWD5-1999-CBP3 carton label: 16 A	published · project-snapshot (unresolved) · carton label — Photographed MWD5-1999-CBP3 carton label: 16 A · Photographed MWD5-1999-CBP3 carton label: 16 A
Ther mosta t rated voltag e, minim um	100 V · published · carton label — Photographed MWD5-1999-CBP3 carton label: 100–240 V AC	published · project-snapshot (unresolved) · carton label — Photographed MWD5-1999-CBP3 carton label: 100–240 V AC · Photographed MWD5-1999-CBP3 carton label: 100–240 V AC
Ther mosta t rated voltag e, maxi mum	240 V · published · carton label — Photographed MWD5-1999-CBP3 carton label: 100–240 V AC	published · project-snapshot (unresolved) · carton label — Photographed MWD5-1999-CBP3 carton label: 100–240 V AC · Photographed MWD5-1999-CBP3 carton label: 100–240 V AC
Plan route length	42.5 m · modelled	modelled · drawn-path · XY projection of the drawn path. Modelled, not published cable length.
Spatia l route length	unknown (unknown; Sampled cable profile. Modelled; unknown until screed levels and cable height resolve. Not published cable length.)	unknown · drawn-path · Sampled cable profile. Modelled; unknown until screed levels and cable height resolve. Not published cable length.
Remai ning confir med produ ct length	unknown (unknown; Balance stays unknown until confirmed product length and the whole spatial profile are both numeric.)	unknown · confirmed product length – modelled spatial route length · product-minus-route · Balance stays unknown until confirmed product length and the whole spatial profile are both numeric.
Ther mosta t printe d IP	IP21 · published	carton label — Photographed MWD5-1999-CBP3 carton label: housing IP21
Entere d minim um spaci ng	unknown	unknown · source not supplied
Entere d cleara nce	100 mm	published · Coldbuster In Screed Heater Kit Installation Manual (July 2023), p. 6: mark 100–200 mm from the walls; Coldbuster recommends 100 mm. Taken as this cable’s manual: it does not name SCK0765L
Cable centre above scree d botto m	unknown	unknown · source not supplied

Selected zones: bathroom

Sampled profile at vertices, plane boundaries and interval midpoints. Cable centre is measured from the bottom face of the selected screed layer (top of the layer below / subfloor stack); not the underside of the tile unless that face is the screed top. Entered screed thickness applies throughout the route; local screed top follows entered finished planes minus layers above. Trade must verify variable thickness, substrate and slope.

Cable section unresolved — missing levels / cable height

Plan route distance (m)	Plan x / y (mm)	Screed bottom / cable / top above datum	Basis or missing
0	150 mm / 2843 mm	unknown / unknown / unknown	plane "Dry area" is unresolved (missing Kano 316 tile-insert waste 120 × 120, brushed nickel, 50 mm outlet level, fall or a control level), cable height above screed bottom, Tiler's screed (heating cable inside) thickness, Tiler's adhesive thickness
0.375	525 mm / 2843 mm	unknown / unknown / unknown	plane "Dry area" is unresolved (missing Kano 316 tile-insert waste 120 × 120, brushed nickel, 50 mm outlet level, fall or a control level), cable height above screed bottom, Tiler's screed (heating cable inside) thickness, Tiler's adhesive thickness
0.75	900 mm / 2843 mm	unknown / unknown / unknown	plane "Dry area" is unresolved (missing Kano 316 tile-insert waste 120 × 120, brushed nickel, 50 mm outlet level, fall or a control level), cable height above screed bottom, Tiler's screed (heating cable inside) thickness, Tiler's adhesive thickness
0.9675	1117.5 mm / 2843 mm	unknown / unknown / unknown	plane "Dry area" is unresolved (missing Kano 316 tile-insert waste 120 × 120, brushed nickel, 50 mm outlet level, fall or a control level), cable height above screed bottom, Tiler's screed (heating cable inside) thickness, Tiler's adhesive thickness
1.185	1335 mm / 2843 mm	unknown / unknown / unknown	plane "Dry area" is unresolved (missing Kano 316 tile-insert waste 120 × 120, brushed nickel, 50 mm outlet level, fall or a control level), cable height above screed bottom, Tiler's screed (heating cable inside) thickness, Tiler's adhesive thickness
1.2295	1335 mm / 2798.5 mm	unknown / unknown / unknown	plane "Dry area" is unresolved (missing Kano 316 tile-insert waste 120 × 120, brushed nickel, 50 mm outlet level, fall or a control level), cable height above screed bottom, Tiler's screed (heating cable inside) thickness, Tiler's adhesive thickness
1.274	1335 mm / 2754 mm	unknown / unknown / unknown	plane "Dry area" is unresolved (missing Kano 316 tile-insert waste 120 × 120, brushed nickel, 50 mm outlet level, fall or a control level), cable height above screed bottom, Tiler's screed (heating cable inside) thickness, Tiler's adhesive thickness
1.4915	1117.5 mm / 2754 mm	unknown / unknown / unknown	plane "Dry area" is unresolved (missing Kano 316 tile-insert waste 120 × 120, brushed nickel, 50 mm outlet level, fall or a control level), cable height above screed bottom, Tiler's screed (heating cable inside) thickness, Tiler's adhesive thickness
1.709	900 mm / 2754 mm	unknown / unknown / unknown	plane "Dry area" is unresolved (missing Kano 316 tile-insert waste 120 × 120, brushed nickel, 50 mm outlet level, fall or a control level), cable height above screed bottom, Tiler's screed (heating cable inside) thickness, Tiler's adhesive thickness
2.084	525 mm / 2754 mm	unknown / unknown / unknown	plane "Dry area" is unresolved (missing Kano 316 tile-insert waste 120 × 120, brushed nickel, 50 mm outlet level, fall or a control level), cable height above screed bottom, Tiler's screed (heating cable inside) thickness, Tiler's adhesive thickness
2.459	150 mm / 2754 mm	unknown / unknown / unknown	plane "Dry area" is unresolved (missing Kano 316 tile-insert waste 120 × 120, brushed nickel, 50 mm outlet level, fall or a control level), cable height above screed bottom, Tiler's screed (heating cable inside) thickness, Tiler's adhesive thickness
2.5035	150 mm / 2709.5 mm	unknown / unknown / unknown	plane "Dry area" is unresolved (missing Kano 316 tile-insert waste 120 × 120, brushed nickel, 50 mm outlet level, fall or a control level), cable height above screed bottom, Tiler's screed (heating cable inside) thickness, Tiler's adhesive thickness
2.548	150 mm / 2665 mm	unknown / unknown / unknown	plane "Dry area" is unresolved (missing Kano 316 tile-insert waste 120 × 120, brushed nickel, 50 mm outlet level, fall or a control level), cable height above screed bottom, Tiler's screed (heating cable inside) thickness, Tiler's adhesive thickness
2.923	525 mm / 2665 mm	unknown / unknown / unknown	plane "Dry area" is unresolved (missing Kano 316 tile-insert waste 120 × 120, brushed nickel, 50 mm outlet level, fall or a control level), cable height above screed bottom, Tiler's screed (heating cable inside) thickness, Tiler's adhesive thickness
3.298	900 mm / 2665 mm	unknown / unknown / unknown	plane "Dry area" is unresolved (missing Kano 316 tile-insert waste 120 × 120, brushed nickel, 50 mm outlet level, fall or a control level), cable height above screed bottom, Tiler's screed (heating cable inside) thickness, Tiler's adhesive thickness
3.5155	1117.5 mm / 2665 mm	unknown / unknown / unknown	plane "Dry area" is unresolved (missing Kano 316 tile-insert waste 120 × 120, brushed nickel, 50 mm outlet level, fall or a control level), cable height above screed bottom, Tiler's screed (heating cable inside) thickness, Tiler's adhesive thickness
3.733	1335 mm / 2665 mm	unknown / unknown / unknown	plane "Dry area" is unresolved (missing Kano 316 tile-insert waste 120 × 120, brushed nickel, 50 mm outlet level, fall or a control level), cable height above screed bottom, Tiler's screed (heating cable inside) thickness, Tiler's adhesive thickness
3.7775	1335 mm / 2620.5 mm	unknown / unknown / unknown	plane "Dry area" is unresolved (missing Kano 316 tile-insert waste 120 × 120, brushed nickel, 50 mm outlet level, fall or a control level), cable height above screed bottom, Tiler's screed (heating cable inside) thickness, Tiler's adhesive thickness
3.822	1335 mm / 2576 mm	unknown / unknown / unknown	plane "Dry area" is unresolved (missing Kano 316 tile-insert waste 120 × 120, brushed nickel, 50 mm outlet level, fall or a control level), cable height above screed bottom, Tiler's screed (heating cable inside) thickness, Tiler's adhesive thickness
4.0395	1117.5 mm / 2576 mm	unknown / unknown / unknown	plane "Dry area" is unresolved (missing Kano 316 tile-insert waste 120 × 120, brushed nickel, 50 mm outlet level, fall or a control level), cable height above screed bottom, Tiler's screed (heating cable inside) thickness, Tiler's adhesive thickness
4.257	900 mm / 2576 mm	unknown / unknown / unknown	plane "Dry area" is unresolved (missing Kano 316 tile-insert waste 120 × 120, brushed nickel, 50 mm outlet level, fall or a control level), cable height above screed bottom, Tiler's screed (heating cable inside) thickness, Tiler's adhesive thickness
4.632	525 mm / 2576 mm	unknown / unknown / unknown	plane "Dry area" is unresolved (missing Kano 316 tile-insert waste 120 × 120, brushed nickel, 50 mm outlet level, fall or a control level), cable height above screed bottom, Tiler's screed (heating cable inside) thickness, Tiler's adhesive thickness
5.007	150 mm / 2576 mm	unknown / unknown / unknown	plane "Dry area" is unresolved (missing Kano 316 tile-insert waste 120 × 120, brushed nickel, 50 mm outlet level, fall or a control level), cable height above screed bottom, Tiler's screed (heating cable inside) thickness, Tiler's adhesive thickness
5.0515	150 mm / 2531.5 mm	unknown / unknown / unknown	plane "Dry area" is unresolved (missing Kano 316 tile-insert waste 120 × 120, brushed nickel, 50 mm outlet level, fall or a control level), cable height above screed bottom, Tiler's screed (heating cable inside) thickness, Tiler's adhesive thickness
5.096	150 mm / 2487 mm	unknown / unknown / unknown	plane "Dry area" is unresolved (missing Kano 316 tile-insert waste 120 × 120, brushed nickel, 50 mm outlet level, fall or a control level), cable height above screed bottom, Tiler's screed (heating cable inside) thickness, Tiler's adhesive thickness
5.471	525 mm / 2487 mm	unknown / unknown / unknown	plane "Dry area" is unresolved (missing Kano 316 tile-insert waste 120 × 120, brushed nickel, 50 mm outlet level, fall or a control level), cable height above screed bottom, Tiler's screed (heating cable inside) thickness, Tiler's adhesive thickness
5.846	900 mm / 2487 mm	unknown / unknown / unknown	plane "Dry area" is unresolved (missing Kano 316 tile-insert waste 120 × 120, brushed nickel, 50 mm outlet level, fall or a control level), cable height above screed bottom, Tiler's screed (heating cable inside) thickness, Tiler's adhesive thickness

39.28	525 mm / 398 mm	unknown / unknown / unknown	plane "Shower" is unresolved (missing Lauxes Next Gen 35 channel 1000 × 100 × 35, brushed nickel, 50 mm outlet WO50-BN level, fall or a control level), cable height above screed bottom, Tiler's screed (heating cable inside) thickness, Tiler's adhesive thickness
39.655	900 mm / 398 mm	unknown / unknown / unknown	plane "Shower" is unresolved (missing Lauxes Next Gen 35 channel 1000 × 100 × 35, brushed nickel, 50 mm outlet WO50-BN level, fall or a control level), cable height above screed bottom, Tiler's screed (heating cable inside) thickness, Tiler's adhesive thickness
39.696	941 mm / 398 mm	unknown / unknown / unknown	plane "Beside the shower (under the bath)" is unresolved (missing Kano 316 tile-insert waste 120 × 120, brushed nickel, 50 mm outlet level, fall or a control level), cable height above screed bottom, Tiler's screed (heating cable inside) thickness, Tiler's adhesive thickness
39.737	982 mm / 398 mm	unknown / unknown / unknown	plane "Beside the shower (under the bath)" is unresolved (missing Kano 316 tile-insert waste 120 × 120, brushed nickel, 50 mm outlet level, fall or a control level), cable height above screed bottom, Tiler's screed (heating cable inside) thickness, Tiler's adhesive thickness
39.7815	982 mm / 353.5 mm	unknown / unknown / unknown	plane "Beside the shower (under the bath)" is unresolved (missing Kano 316 tile-insert waste 120 × 120, brushed nickel, 50 mm outlet level, fall or a control level), cable height above screed bottom, Tiler's screed (heating cable inside) thickness, Tiler's adhesive thickness
39.826	982 mm / 309 mm	unknown / unknown / unknown	plane "Beside the shower (under the bath)" is unresolved (missing Kano 316 tile-insert waste 120 × 120, brushed nickel, 50 mm outlet level, fall or a control level), cable height above screed bottom, Tiler's screed (heating cable inside) thickness, Tiler's adhesive thickness
39.867	941 mm / 309 mm	unknown / unknown / unknown	plane "Beside the shower (under the bath)" is unresolved (missing Kano 316 tile-insert waste 120 × 120, brushed nickel, 50 mm outlet level, fall or a control level), cable height above screed bottom, Tiler's screed (heating cable inside) thickness, Tiler's adhesive thickness
39.908	900 mm / 309 mm	unknown / unknown / unknown	plane "Shower" is unresolved (missing Lauxes Next Gen 35 channel 1000 × 100 × 35, brushed nickel, 50 mm outlet WO50-BN level, fall or a control level), cable height above screed bottom, Tiler's screed (heating cable inside) thickness, Tiler's adhesive thickness
40.283	525 mm / 309 mm	unknown / unknown / unknown	plane "Shower" is unresolved (missing Lauxes Next Gen 35 channel 1000 × 100 × 35, brushed nickel, 50 mm outlet WO50-BN level, fall or a control level), cable height above screed bottom, Tiler's screed (heating cable inside) thickness, Tiler's adhesive thickness
40.658	150 mm / 309 mm	unknown / unknown / unknown	plane "Shower" is unresolved (missing Lauxes Next Gen 35 channel 1000 × 100 × 35, brushed nickel, 50 mm outlet WO50-BN level, fall or a control level), cable height above screed bottom, Tiler's screed (heating cable inside) thickness, Tiler's adhesive thickness
40.7025	150 mm / 264.5 mm	unknown / unknown / unknown	plane "Shower" is unresolved (missing Lauxes Next Gen 35 channel 1000 × 100 × 35, brushed nickel, 50 mm outlet WO50-BN level, fall or a control level), cable height above screed bottom, Tiler's screed (heating cable inside) thickness, Tiler's adhesive thickness
40.747	150 mm / 220 mm	unknown / unknown / unknown	plane "Shower" is unresolved (missing Lauxes Next Gen 35 channel 1000 × 100 × 35, brushed nickel, 50 mm outlet WO50-BN level, fall or a control level), cable height above screed bottom, Tiler's screed (heating cable inside) thickness, Tiler's adhesive thickness
41.122	525 mm / 220 mm	unknown / unknown / unknown	plane "Shower" is unresolved (missing Lauxes Next Gen 35 channel 1000 × 100 × 35, brushed nickel, 50 mm outlet WO50-BN level, fall or a control level), cable height above screed bottom, Tiler's screed (heating cable inside) thickness, Tiler's adhesive thickness
41.497	900 mm / 220 mm	unknown / unknown / unknown	plane "Shower" is unresolved (missing Lauxes Next Gen 35 channel 1000 × 100 × 35, brushed nickel, 50 mm outlet WO50-BN level, fall or a control level), cable height above screed bottom, Tiler's screed (heating cable inside) thickness, Tiler's adhesive thickness
41.538	941 mm / 220 mm	unknown / unknown / unknown	plane "Beside the shower (under the bath)" is unresolved (missing Kano 316 tile-insert waste 120 × 120, brushed nickel, 50 mm outlet level, fall or a control level), cable height above screed bottom, Tiler's screed (heating cable inside) thickness, Tiler's adhesive thickness
41.579	982 mm / 220 mm	unknown / unknown / unknown	plane "Beside the shower (under the bath)" is unresolved (missing Kano 316 tile-insert waste 120 × 120, brushed nickel, 50 mm outlet level, fall or a control level), cable height above screed bottom, Tiler's screed (heating cable inside) thickness, Tiler's adhesive thickness
41.6235	982 mm / 175.5 mm	unknown / unknown / unknown	plane "Beside the shower (under the bath)" is unresolved (missing Kano 316 tile-insert waste 120 × 120, brushed nickel, 50 mm outlet level, fall or a control level), cable height above screed bottom, Tiler's screed (heating cable inside) thickness, Tiler's adhesive thickness
41.668	982 mm / 131 mm	unknown / unknown / unknown	plane "Beside the shower (under the bath)" is unresolved (missing Kano 316 tile-insert waste 120 × 120, brushed nickel, 50 mm outlet level, fall or a control level), cable height above screed bottom, Tiler's screed (heating cable inside) thickness, Tiler's adhesive thickness
41.709	941 mm / 131 mm	unknown / unknown / unknown	plane "Beside the shower (under the bath)" is unresolved (missing Kano 316 tile-insert waste 120 × 120, brushed nickel, 50 mm outlet level, fall or a control level), cable height above screed bottom, Tiler's screed (heating cable inside) thickness, Tiler's adhesive thickness
41.75	900 mm / 131 mm	unknown / unknown / unknown	plane "Shower" is unresolved (missing Lauxes Next Gen 35 channel 1000 × 100 × 35, brushed nickel, 50 mm outlet WO50-BN level, fall or a control level), cable height above screed bottom, Tiler's screed (heating cable inside) thickness, Tiler's adhesive thickness
42.125	525 mm / 131 mm	unknown / unknown / unknown	plane "Shower" is unresolved (missing Lauxes Next Gen 35 channel 1000 × 100 × 35, brushed nickel, 50 mm outlet WO50-BN level, fall or a control level), cable height above screed bottom, Tiler's screed (heating cable inside) thickness, Tiler's adhesive thickness
42.5	150 mm / 131 mm	unknown / unknown / unknown	plane "Shower" is unresolved (missing Lauxes Next Gen 35 channel 1000 × 100 × 35, brushed nickel, 50 mm outlet WO50-BN level, fall or a control level), cable height above screed bottom, Tiler's screed (heating cable inside) thickness, Tiler's adhesive thickness

Keep-out	x / y / width / height (mm)	Source
Bath	1082 mm / 0 mm / 1028 mm / 1018 mm	Owner, 7 Oct 2026: floor under the bath left unheated so the heated area stays within the 5.1 m² the 42.5 m cable covers
Toilet	1435 mm / 2409.5 mm / 675 mm / 381 mm	Owner, 8 Oct 2026: no cable under the toilet (its pan is screwed to the floor through brackets); cable stays under the vanity

- warning · heating_signoff: Proposed route only: manufacturer and licensed electrician must review cable identity, length, output, bend radius, spacing, exclusions, cover, sensor, cold tails, waterproofing and electrical installation. No compliance approval. No electrical approval is implied.
- warning · heating_product_unresolved: sample-sck0765l-carton is not an accepted heating-cable in this library. Checks use the travelling snapshot only when its productId is that id or in its series.
- warning · heating_product_unresolved: sample-mwd5-1999-cbp3 is not an accepted thermostat in this library. Checks use the travelling snapshot only when its productId is that id or in its series.
- warning · heating_metadata_unknown: Unknown cable/product information: manufacturer, minSpacing, depthFromBottom.
- warning · heating_route_length_unknown: Spatial cable length and confirmed product-length balance remain unknown until the whole route has resolved screed levels and cable height; the plan projection is modelled, not cable length.
- warning · heating_ip_location: Thermostat printed IP21 (published, carton label — Photographed MWD5-1999-CBP3 carton label: housing IP21) vs planned location outside a wet room (Hallway wall outside the bathroom, next to the light switch (right as you look into the bathroom), 1000 mm off the floor. Not drawn on this plan.; Owner, 5 Oct 2026; height raised to 1000 mm, 7 Oct 2026). This check does not decide zone suitability; the electrician decides. No compliance approval.
- warning · heating_depth_unknown: Cable section remains unresolved until screed layer, floor levels and cable height (bottom face of the selected screed layer (top of the layer below / subfloor stack); not the underside of the tile unless that face is the screed top) are entered.